

Questions with Answer Keys

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Q1 - Single Correct

The equation of circle of minimum radius which touches both the parabolas $y = x^2 + 2x + 4$ and $x = y^2 + 2y + 4$ is

(1) $4x^2 + 4y^2 - 11x - 11y - 31 = 0$

(2) $4x^2 + 4y^2 - 11x - 11y - 13 = 0$

(3) $4x^2 + 4y^2 - 11x - 11y + 11 = 0$

(4) $4x^2 + 4y^2 - 11x - 11y - 6 = 0$

Q2 - Single Correct

If the parabola $y = ax^2 + bx + c$ has vertex $(4, 2)$ and $a \in [1, 3]$, then difference between the extreme values of abc is equal to

(1) 3600

(2) 144

(3) 3456

(4) Can't found

Q3 - Single Correct

The line $x - b + \lambda y = 0$ cuts the parabola $y^2 = 4ax$ ($a > 0$) at $P(t_1)$ and $Q(t_2)$. If $b \in [2a, 4a]$, then the range of $t_1 t_2$, where $\lambda \in R$ is

(1) $[-4, -2]$

(2) $[2, 4]$

(3) $[4, 16]$

(4) $[-16, -4]$

Q4 - Single Correct

Consider one side AB of a square $ABCD$ on the line $y = 2x - 17$ and the other two vertices C, D lies on the parabola $y = x^2$, then the maximum possible area of the square $ABCD$ (in sq units) is

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(1) 160

(2) 1280

(3) 640

(4) 320

Q5 - Single Correct

The normals are drawn from $(2k, 0)$ to the curve $y^2 = 4x$. One normal is X -axis. Then, the value of k for which other two normals are perpendicular to each other, is

(1) $3/2$

(2) 2

(3) $5/2$ (4) $7/2$

Q6 - Single Correct

Two unequal parabolas have the same common axis which is X -axis and have the same vertex which is the origin with their concavities in opposite direction. If a variable line parallel to the common axis meet the parabolas in P and P' , then the locus of middle point of PP' is

(1) a parabola

(2) a circle

(3) an ellipse

(4) a hyperbola

Q7 - Single Correct

The locus of the centre of a circle, which intercepts a chord of given length $2a$ on the X -axis and passes through a given point on the Y -axis distant b from the origin, is the curve

(1) $x^2 - 2yb + b^2 = a^2$ (2) $x^2 - 2ay + b^2 = a^2$

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(3) $y^2 - 2ax + a^2 = b^2$

(4) $y^2 - 2bx + b^2 = a^2$

Q8 - Single Correct

A variable parabola is drawn to pass through A and B , the ends of a diameter of a given circle with centre at the origin and radius c and to have as directrix a tangent to a concentric circle of radius a ($a > c$); the axes being AB and a perpendicular diameter. then the locus of focus of parabola is

(1) a circle

(2) a parabola

(3) an ellipse

(4) a hyperbola

Q9 - Single Correct

Let A and B be two points on a parabola $y^2 = x$ with vertex V such that VA is perpendicular to VB and θ is the angle between the chord VA and the axis of the parabola. The value of $\left| \frac{VA}{VB} \right|$ is

(1) $\tan \theta$ (2) $\tan^3 \theta$ (3) $\cot^2 \theta$ (4) $\cot^3 \theta$

Q10 - Single Correct

C is the centre of the circle with centre $(0,1)$ and radius unity. P is the parabola $y = ax^2$. The set of values of a for which they meet at a point other than the origin is

(1) $a > 0$ (2) $a \in \left(0, \frac{1}{2}\right)$ (3) $\left(\frac{1}{4}, \frac{1}{2}\right)$ (4) $\left(\frac{1}{2}, \infty\right)$

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Q11 - Single Correct

TP and TQ are tangents to the parabola $y^2 = 4ax$ at P and Q . If the chord PQ passes through the fixed point $(-a, b)$, then the locus of T is

(1) $ay = 2b(x - b)$

(2) $bx = 2a(y - a)$

(3) $by = 2a(x - a)$

(4) $ax = 2b(y - b)$

Q12 - Single Correct

Through the vertex O of the parabola $y^2 = 4ax$, two chords OP and OQ are drawn and the circles on OP and OQ as diameters intersect in R . If θ_1, θ_2 and ϕ are the angles made with the axes by the tangents P and Q on the parabola and by OR , then the value of $\cot \theta_1 + \cot \theta_2$ is

(1) $-2 \tan \phi$

(2) $-2 \tan(\pi - \phi)$

(3) 0

(4) $2 \cot \phi$

Q13 - Multiple Correct

The locus of mid-point of the focal radii of a variable point moving on the parabola $y^2 = 8x$ is a parabola whose

(1) latusrectum is half the latusrectum of original parabola

(2) vertex is $(1, 0)$ (3) directrix is Y -axis(4) focus has the coordinates $(2, 0)$

Q14 - Multiple Correct

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P is the point on the parabola $y^2 = 4ax$ ($a > 0$) whose vertex is A . PA is produced to meet the directrix in D and M is the foot of perpendicular from P on the directrix. If a circle is described on MD as a diameter, then its intersect the X -axis at a point whose coordinates are

- (1) $(-3a, 0)$
- (2) $(-a, 0)$
- (3) $(-2a, 0)$
- (4) $(a, 0)$

Q15 - Multiple Correct

A variable circle is described to pass through the point $(1, 0)$ and tangent to the curve $y = \tan(\tan^{-1} x)$. The locus of centre of the circle is a parabola whose

- (1) length of the latusrectum is $2\sqrt{2}$
- (2) axis of symmetry has the equation $x + y = 1$
- (3) vertex has the coordinates $(3/4, 1/4)$
- (4) directrix is $x - y = 0$

Q16 - Multiple Correct

Let A and B be two distinct points on the parabola $y^2 = 4x$. If axis of the parabola touches a circle of radius r having AB as its diameter, then the slope of the line joining A and B can be

- (1) $-1/r$
- (2) $1/r$
- (3) $2/r$
- (4) $-2/r$

Q17 - Multiple Correct

The equation of a tangent to the parabola $y^2 = 12x$ which make an angle 45° with the $y = 3x + 77$, is

- (1) $2x - 4y + 3 = 0$

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(2) $x - 2y + 12 = 0$

(3) $4x + 2y + 3 = 0$

(4) $2x + y - 12 = 0$

Q18 - Multiple Correct

The focus of the parabola is (1,1) and the tangent at the vertex has the equation $x + y = 1$. Then

(1) the equation of parabola is $(y - x)^2 = 2(x + y - 1)$

(2) the equation of parabola is $(x - y)^2 = 4(x + y - 1)$

(3) the coordinates of vertex are $(1/2, 1/2)$

(4) the length of latus rectum is $2\sqrt{2}$

Q19 - Multiple Correct

P is a point on the parabola $y^2 = 4x$, where abscissa and ordinate are equal. Then, the equation of circle passing through the focus and touching the parabola at P is

(1) $x^2 + y^2 + 13x + 2y + 12 = 0$

(2) $x^2 + y^2 - 3x - 18y + 2 = 0$

(3) $x^2 + y^2 + 13x - 2y - 12 = 0$

(4) $x^2 + y^2 + x = 0$

Q20 - Multiple Correct

The equation of common tangent(s) to the parabola $y = x^2$ and $y = -x^2 + 4x - 4$ is/are

(1) $y = 4(x - 1)$

(2) $y = 0$

(3) $y = -4(x - 1)$

(4) $y = -30x - 50$

Q21 - Multiple Correct

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Normals are drawn at points A, B and C on the parabola $y^2 = 4x$ which intersect at $P(h, k)$. The locus of the point $P(h, k)$, is the line joining two of the point having slope equal to 2

(1) $x + y = 1$

(2) $x - y = 3$

(3) $y^2 = 2(x - 1)$

(4) $y^2 = 2\left(x - \frac{1}{2}\right)$

Q22 - Multiple Correct

$ABCD$ and $EFGC$ are squares and the curve $y = k\sqrt{x}$ passes through the origin D and the points B and F .

The ratio $\frac{FG}{BC}$ is

(1) $\frac{\sqrt{5}+1}{2}$

(2) $\frac{\sqrt{3}+1}{2}$

(3) $\frac{\sqrt{5}+1}{4}$

(4) $\frac{\sqrt{3}+1}{4}$

Q23 - Paragraph 1

Passage I (For Question 23, 24)

Tangents are drawn to the parabola $y^2 = 4x$ at the point P which is the upper end of latusrectum.

Image of the parabola $y^2 = 4x$ in the tangent line at the point P is

(1) $(x + 4)^2 = 16y$

(2) $(x + 2)^2 = 8(y - 2)$

(3) $(x + 1)^2 = 4(y - 1)$

(4) $(x - 2)^2 = 2(y - 2)$

Q24 - Paragraph 1

The radius of the circle touching the parabola $y^2 = 4x$ at the point P and passing through its focus is

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- (1) 1
- (2) $\sqrt{2}$
- (3) $\sqrt{3}$
- (4) 2

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Q25 - Paragraph 2

Passage II (For Question 25, 26)

$y = f(x)$ is a parabola of the form $y = x^2 + ax + 1$. Its tangent at the point of intersection of Y-axis and the parabola also touch the circle $x^2 + y^2 = r^2$. It is known that no points of the parabola is below X-axis.

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The radius of the circle where a attains its maximum value, is

- (1) $1/\sqrt{10}$
- (2) $1/\sqrt{5}$
- (3) 1
- (4) $\sqrt{5}$

Q26 - Paragraph 2

The slope of the tangent when the radius of the circle is maximum, is

- (1) 0
- (2) 1
- (3) -1
- (4) not defined

Q27 - Paragraph 3

Passage III (For Question 27, 28)

Let t be a positive number. Draw two tangents from the point $(t, -1)$ to the parabola $y = x^2$. If the area bounded by the tangent lines and the parabola is denoted by $S(t)$.

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 $S(t)$ is given by

(1) $\frac{2}{3}(1+t^2)^{3/2}$

(2) $\frac{1}{3}(1+t^2)^{3/2}$

(3) $\frac{3}{4}(1+t^2)^{3/2}$

(4) $\frac{2}{3}(1-t^2)^{2/3}$

Q28 - Paragraph 3

 $\frac{S(t)}{\sqrt{t}}$ is minimum, then t is equal to

(1) $1/\sqrt{5}$

(2) $\sqrt{5}$

(3) $2/\sqrt{5}$

(4) $1/\sqrt{3}$

Q29 - Paragraph 4

Passage IV (For Question 29, 30)

Consider two parabolas $C_1 : (x-1)^2 + f(k)$ and $C_2 : x^2 - 2x + 4y - 7 = 0$, where

$$f(k) = |k^2 - 3k + 2| + \frac{1}{4}, k \in R$$

If C_1 and C_2 does not intersect, then $|k^2 - 3k + 2|$ is greater than

(1) $1/4$

(2) $3/4$

(3) $7/4$

(4) 1

Q30 - Paragraph 4

The distance between the foci of C_1 and C_2 for which $f(k)$ is least, is

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(1) $3/2$

(2) $1/4$

(3) $3/4$

(4) $1/2$

Q31 - Paragraph 5

Passage V (For Question 31, 32)

A point P is moving in XY -plane such that if it lies inside the triangle formed by $|x| = 3$ and $y = \pm x$, then its distance from the origin is always r , otherwise it always lies on the curve $y^2 = 4|x|$

The range of r such that there exist exactly eight points on locus P from which if we draw two tangents on the parabolic part, then they are at right angle, is

(1) $r > 1$

(2) $1 < r < \sqrt{2}$

(3) $1 < r \leq \sqrt{2}$

(4) $r \geq 1$

Q32 - Paragraph 5

If $r = 1$, then the radius of the largest circle which touches the locus of P at three points is given by

(1) $(x - 5)^2 + y^2 = 16$

(2) $(x + 5)^2 + y^2 = 0$

(3) $x^2 + (y - 5)^2 = 16$

(4) $\left(x - \frac{1}{2}\right)^2 + y^2 = \frac{1}{4}$

Q33 - Integer Type

The orthocentre of any triangle formed by three tangents to a parabola $y^2 = 4x$, lies on $x = \lambda$, then the value of $|\lambda|$ is

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Q34 - Integer Type

Two straight lines are perpendicular to each other. One of them touches the parabola $y^2 = 4a(x + a)$ and the other touches the parabola $y^2 = 4b(x + b)$. If the point of intersection of the lines lie on $x + \lambda a + \mu b = 0$, then the value of $(\lambda + \mu)$ is

Q35 - Integer Type

If the parabolas $y^2 = \lambda x$ and $25[(x - 3)^2 + (y + 2)^2] = (3x - 4y - 2)^2$ are equal, then the value of λ is

Q36 - Integer Type

PQ is a focal chord of $y^2 = 4ax$, PP' and QQ' are normal chord at P and Q respectively. Then, the value of $P'Q'/PQ$ is

Q37 - Integer Type

The locus of point from which three normals drawn to the curve $y^2 + 2y - 4x + 5 = 0$ are such that two of them are perpendicular is $\left(y - \frac{\mu}{2}\right)^2 - (x - \lambda) = 0$, then the value of $(\lambda - 2\mu)$ is

Q38 - Integer Type

From a point T is tangent drawn to touch the parabola $y^2 = 16x$ at $P(16, 16)$. If S is the focus of the parabola and $\angle TPS = \cot^{-1}(k)$, then the value of k is

Q39 - Integer Type

Radius of the largest circle which passes through the focus of the parabola $y^2 = 4x$ and contained in it, is

Q40 - Integer Type

Normals are drawn from the point P with slopes $m_1 \cdot m_2 = \alpha$ in a part of the parabola itself, then the value of α is

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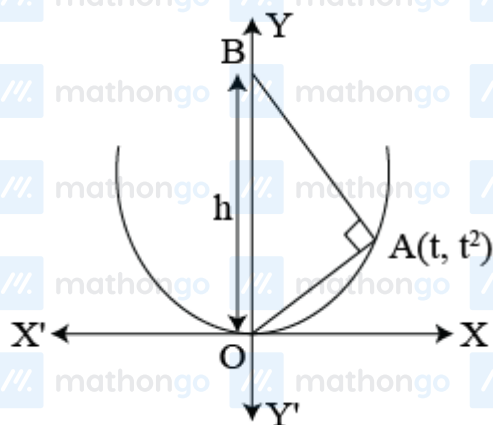
Q41 - Integer Type

The tangent at the point $P(1, 1)$ on the curve $y = x^2 + bx - b$ meets the positive X and Y -axes at A and B respectively and O is the origin. If b_1, b_2, b_3 are the values of b for which the area of the ΔOAB is 2 sq units with $b_1 < b_2 < b_3$, then $|b_1| + |b_2 + b_3|$ is

Q42 - Integer Type

The figure shown a right angle with its hypotenuse OB along the Y -axis and its vertex A on the parabola $y = x^2$. Let h represents the length of the hypotenuse which depends on the x -coordinate of the point A .

Then, $\lim_{x \rightarrow 0} (t^2 + 1)$ is



Q43 - Integer Type

The mid-point of the chord intercepted on the line $4x - 3y + 4 = 0$ by the parabola $y^2 = 8x$ is (λ, μ) , then the value of $(4\lambda + \mu)$ is

Q44 - Integer Type

The polar of a point with respect to $y^2 = 4x$ touches $x^2 = 4y$. If the locus of this point is $xy = \lambda$, then the value of $|\lambda|$ is

Q45 - Integer Type

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The locus of the poles of tangents to the parabola $y^2 = 4ax$ with respect to the parabola $y^2 = 4bx$ is parabola $y^2 = \frac{\lambda b^2 x}{a}$, then the value of λ is

Q46 - Integer Type

Three normals are drawn from $(2k, 0)$ to the parabola $y^2 = 4x$ ($k > 1$). One normal is always the X -axis.

Then, the value of $(2k)$ for which other two normals are perpendicular to each other, is

Q47 - Integer Type

The locus of the points such that the sum of the angles which three normals drawn from them to a parabola make with the axis of the parabola $y^2 = 4ax$ is a constant λ , is $2x + (\cot \lambda)y - \mu a = 0$. Then, the value of μ is

Q48 - Integer Type

A normal drawn at a point P on the parabola $y^2 = 4ax$ meets the curve again at Q . If the least distance of the point Q from the axis of the parabola is $(\lambda\sqrt{2}a)$, then the value of λ is

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